

Riphah International University Islamabad

Gulberg Green Campus



Programming Fundamentals

The Quizz Master Pro

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Date of Submission: 26/ 12/ 2023

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Project Proposal

1. INTRODUCTION

"Quiz Master Pro" is a console-based interactive quiz game developed in C++. The project demonstrates fundamental programming concepts including CRUD operations, loops, conditional statements, and functions. Users can play quizzes on three different topics with varying difficulty levels, while administrators can manage the question database by adding, viewing, updating, and deleting questions.

2. PROBLEM STATEMENT / OBJECTIVE

The objective of this project is to create an engaging quiz game that:

- 1. Implements CRUD operations (Create, Read, Update, Delete) for question management
- 2. Uses loops for repetitive tasks (menu display, question iteration)
- 3. Uses conditional statements for decision making (if-else, switch)
- 4. Provides an interactive gaming experience with scoring system
- 5. Demonstrates modular programming using functions
- 6. Includes password protection for secure access

3. TECHNICAL REQUIREMENTS

Requirement	Implementation in Project
File Management	NOT USED (as required)
CRUD Operations	Add, Display, Update, Delete questions
Loops	for, while, do-while (menu, questions, options)
Conditional Statements	if-else, switch-case (game logic, menu)

Requirement	Implementation in Project
Functions	7+ functions (modular approach)
Structures	Question structure
Arrays	Question bank array

4. FEATURES OF THE SYSTEM

4.1 Admin Features (CRUD)

Operation	Description
C - Create	Add new questions with statement, type, options, answer
R - Read	Display all stored questions with details
U - Update	Modify existing questions completely
D - Delete	Remove specific questions by index number

4.2 Player Features (Game)

Category	Options Available
Topics	1. Riddles, 2. Historical Figures, 3. Animal World
Difficulty	1. Easy, 2. Medium, 3. Difficult
Question Type	1. True/False, 2. Multiple Choice
Scoring	+1 per correct answer, total score at end
Questions per game	5 questions per session
Feedback	Immediate correct/wrong response

4.3 Security Features

1. Password-protected login (Password: 23236)
2. Maximum 3 login attempts

5. HOW LOOPING STRUCTURES ARE USED

Loop Type	Where Used	Purpose
do-while	Main Menu	Game runs until user chooses exit (option 6)
do-while	Login System	3 attempts for password entry
while	Add Questions	Continue adding until user says 'n'
for	Display Questions	Iterate through question array
for	MCQ Options	Loop through 4 options (A, B, C, D)
for	Delete Question	Shift array elements after deletion

6. HOW CONDITIONAL STATEMENTS ARE USED

Statement	Where Used	Purpose
if-else	Login	Check if password matches
if-else	Answer checking	Determine correct/incorrect
if-else	Question type	T/F vs MCQ handling
if-else	Score calculation	Increment score on correct answer
switch-case	Main Menu	Navigate through options
nested if	Game selection	Topic × Mode × Type combinations

7. FUNCTIONS USED (Modular Programming)

Function Name	Purpose
ourdata()	Display project information and team members
menu()	Display main menu options
addQuestion()	CREATE operation - add new question
displayQuestion()	READ operation - view all questions
updateQuestion()	UPDATE operation - modify existing question
deleteQuestion()	DELETE operation - remove question
playgame()	Main game logic with scoring

8. SAMPLE TEST CASES

Test Case	Input	Expected Output
Login with correct password	23236	Access granted
Login with wrong password (3 times)	1111, 2222, 3333	Access denied
Add new question	Statement, type, options, answer	"Question added successfully"
Play game - Easy Riddles T/F	Topic 1, Mode 1, Type 1	5 riddles questions
Correct answer to MCQ	'A'	"Correct!! +1 score"
Delete question	Index number	"Deleted successfully"

9. LIMITATIONS

1. No file storage - questions lost when program closes
2. Maximum 100 questions (array size limit)

3. Console-based interface (no GUI)
4. Single player only
5. No timer for questions
6. No hint system
7. Questions are hardcoded (not dynamic)

Note: File storage was NOT required as per assignment, so this is acceptable.

10. FUTURE IMPROVEMENTS

1. Add file handling to save questions permanently
2. Create GUI version using graphics library
3. Add multiplayer mode
4. Add timer for each question
5. Add lifelines (50-50, audience poll)
6. Add sound effects
7. Store high scores
8. Add more topics and questions

11. CONCLUSION

The Quiz Master Pro successfully fulfills all assignment requirements: CRUD operations implemented (Add, Display, Update, Delete), Loops used (for, while, do-while), Conditional statements used (if-else, switch), Creative game design with 3 topics, 3 difficulties, 2 question types, Password protection for security, Modular code with functions. The project demonstrates a strong understanding of Programming Fundamentals concepts and provides an engaging user experience.

12. DECLARATION

We hereby declare that this project is our original work and has not been submitted for any other course or assignment.

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